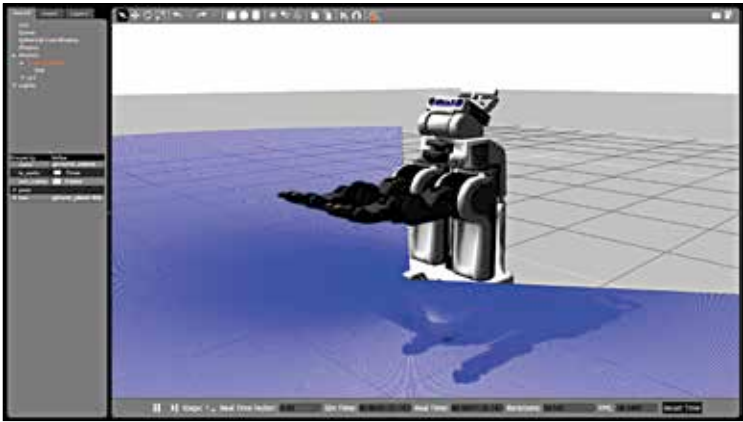




Credit: O'Reilly

ROS Simulator

a robot that can run and act like the actual thing. Once the simulation is complete, your computer will offer your programmes with the same ROS API as if you were in the computer of a real robot. That's precisely the same as with the ROS bags, with the added benefit that you can now send commands to the robot, and the simulated robot will respond appropriately. That's fantastic!

ROS comes pre-installed with the Gazebo simulator (<http://gazebo.sim.org/>) for robot simulations. All you need is a running simulation of the robot you want to programme. Typically, the robot's developers include a simulation of their creation, which you can download and execute on the Gazebo simulator.

Installing simulations and getting them to operate may be more difficult. If you want to skip all of that, we recommend using ROS Development Studio, which includes simulations that can be run with a single click as well as ROS, an IDE, and other useful tools.

Debugging

You'll have to debug the code as well. The majority of the IDEs we've seen contain built-in debugging capabilities. You can, however, use any of the normal debugging tools for C++ programmes, including but not limited to:

- GDB (<http://sourceware.org/gdb/>)
- Oprofile (<http://oprofile.sourceforge.net/news/>)
- Valgrind (<http://valgrind.org/>) 